

NB01: Introduction to Language Processing

Language Processing

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✓ 1.1 Language Processing

Computer languages

- A **computer language** is any formal language used to communicate with a computer
- It is defined by its:
 - an *alphabet*: symbols from which input strings are built
 - *syntax*: rules that determine what are valid input strings
 - *semantics*: rules that assign precise, unambiguous meaning to input strings
- They differ from *natural languages* by having strict syntax and unambiguous semantics

Families of computer languages

Computer languages come in many shapes and goals:

- **Programming languages**: C, Java, Haskell, Python, ...
- **Markup languages**: HTML, TeX, Markdown, ...
- **Data exchange formats**: JSON, YAML, XML, ...
- **Modelling languages**: UML, SysML, BPMN, ...
- **Specification languages**: OCL, Alloy, JML, ...
- **Query languages**: SQL, XQuery, SPARQL, ...

They can also be classified by their scope:

- **General-purpose languages** (GPL): defined to be used in all contexts

- **Domain-specific languages** (DSL): specific to a particular application

Language processing

- **Language processing** is making a machine doing something with a structured text written in a computer language
- It involves turning *text* into a structured **intermediate representation** (IR) and then acting on it
- The main goal is to translate the input language into an output representation

Example: Programming languages

- Programs are written as *plain text*
- The alphabet is the set of allowed characters
 - Traditionally ASCII, increasingly Unicode
- Not all strings over the alphabet are valid programs
 - Each language defines its own syntax rules
- Programming languages have rich syntax and complex semantics, making them a quintessential target of language processing

Compilable vs. Interpreted languages

- Different approaches to language execution
 - Describe *how* programs are processed, not what the language *is*
- **Compilation**
 - Program is translated into another (usually lower-level) language
 - Execution is delegated to a runtime or hardware
 - Compilation can be computationally expensive, but enables global analysis and optimization
 - Typically targets a specific platform or architecture, which may reduce portability
- **Interpretation**
 - Program is executed directly by an interpreter
 - Code is processed incrementally / on-the-fly

- Usually less efficient than compiled code, fewer opportunities for global optimization
- More portable, only requires an interpreter on the target machine
- Many languages *combine* both models:
 - Compile to an intermediate representation
 - Then interpret it or compile it just-in-time (JIT)
 - Examples:
 - Python (bytecode + virtual machine)
 - Java (bytecode + JVM)
 - JavaScript (JIT compilation)

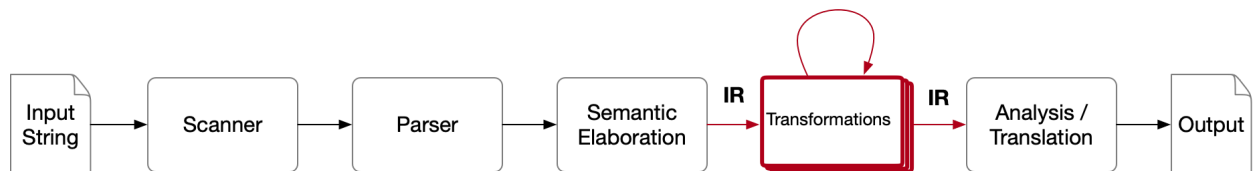
Example: Markup & Data languages

- Not all formal languages are executable
 - Some exist to represent structured information, rather than to be run
- **Markup languages**
 - Annotate text to convey structure or presentation, often also human-readable
 - Examples: HTML, Markdown
- **Data / Interchange languages**
 - Encode structured data for machine processing
 - Examples: XML, JSON, YAML
- Processing these languages involves converting between formats or generating derived outputs
- Configuration files, document generation, and system interoperability all rely on these languages

Language processing pipeline

- Language processing is a complex task, and is organized as *pipeline*
- Each phase adds structure and reduces ambiguity, implementing classic separation of concerns
 - **Lexical analysis**: split the string into tokens
 - **Syntactic analysis**: check structure against a grammar, build a tree
 - **Semantic elaboration**: enforce rules beyond syntax (types, scopes, consistency)

- **Transformation:** convert into another representation (may happen multiple times)
- **Output generation:** produce a result, depending on the goal of the process (compilation, transpilation, interpretation, analysis, visualization, ...)
- In each phase the user must specify what is valid for the language being processed
- Each phase has specific tool support



Example: Python source code

- Input text

```
for i in range(10): print(i)
```

- Tokens, after lexical analysis

```
(FOR, (for)), (identifier, (i)), (IN, (in)), (identifier, (range)), (LEFTPAREN, ()),
(number, (10)), (RIGHTPAREN, ()), (COLON, ()), (identifier, (print)), (LEFTPAREN,
()), (identifier, (i)), (RIGHTPAREN, ())
```

- Syntax tree, after syntactical analysis

```

for-loop
|__ identifier (i)
|__ expression
|   |__ function-call
|       |__ identifier (range)
|       |__ arguments
|           |__ number (10)
|__ block
    |__ function-call
        |__ identifier (print)
        |__ arguments
            |__ identifier (i)
  
```

- Semantics, attach meaning to the syntax tree

```

for-loop
|__ identifier (i)           [bind i : int]
|__ expression              [evaluates to iterable]
|   |__ function-call
|       |__ identifier (range) [builtin : int → iterable[int]]
|       |__ arguments
|           |__ number (10)    [value = 10]
|__ block                   [executed once per element]
    |__ function-call
        |__ identifier (print) [side effect: output]
        |__ arguments
            |__ identifier (i) [value changes each iteration]

```

✓ What errors can it catch?

- Lexical errors

```
for 0xy in range(10): print(0xy)
```

✗ In Python, no token starts by numbers and is followed by alphabetic symbols

- Syntactic errors

```
for i in range(10) { print(i) }
```

✗ In Python, blocks cannot be defined with brackets

- Semantic errors

```
for i in range(10): print(x)
```

✗ In Python, undefined variables are caught in static time

```
for i in range(10): print("value: " + i)
```

! But Python does not perform static type checking, only in runtime

```
for i in range(10): print("value: " + i)
```

```

-----
TypeError                                Traceback (most recent call last)
/tmp/ipython-input-1811290542.py in <cell line: 0>()
----> 1 for i in range(10): print("value: " + i)

TypeError: can only concatenate str (not "int") to str

```

✓ 1.2 Formal Languages

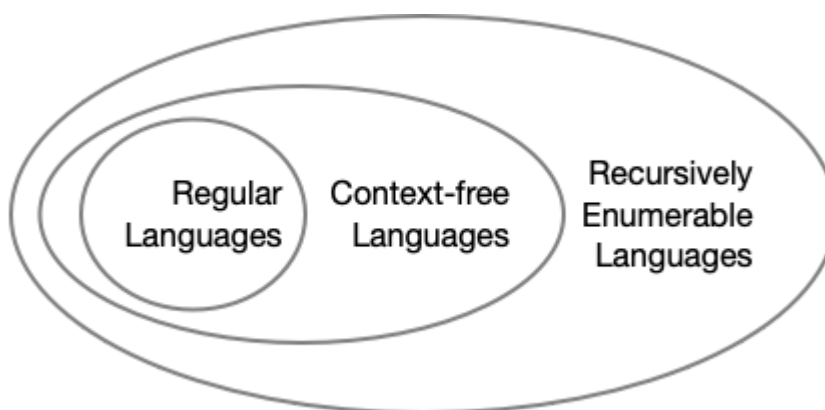
- Computer languages are instances of a general concept: **formal languages**
- A formal language is simply a *set of words* over a finite alphabet
- Words must follow well-defined rules (syntax)
- Studying formal languages allows us to:
 - Predict what can be recognized automatically
 - Design compilers, interpreters, and analyzers
 - Understand limits of what computers can do

Definition

- A formal language L is just a set of *words* (or *strings*), sequences of *symbols* drawn from an *alphabet* Σ
- Words $x, y, z, \dots \in \Sigma^*$ are finite sequences of symbols $a, b, c, \dots \in \Sigma$
 - The length of a word is given by $|w|$
 - The empty word (length 0) is denoted by ε
- A language L is a subset of all possible words, $L \subseteq \Sigma^*$
- The focus is on language *recognition*, how to decide whether a word belongs to the language
 - This is an abstract version of a language processor
- Theory of computation analyses how certain operations can be performed for different classes of languages, and what are their computational limits
 - Does a word w belong to a language L ?
 - Are two languages L_1 and L_2 equivalent, i.e., contain the same words?
 - How does the union $\cup$ or intersection $\cap$ of two languages L_1 and L_2 affect their properties?
- The answer will depend on the class of languages being considered

Chomsky hierarchy of formal languages

- Languages are classified by structure and expressive power
- Recognition / processing tools, as well as limits to efficiency, vary depending on the class
- Type 3 – Regular
 - Patterns without nested dependencies
 - Recognized by *finite automata* or *regex engines*
 - Examples: identifiers, numbers, simple commands
- Type 2 – Context-Free
 - Nested, hierarchical structures
 - Recognized by *pushdown automata* or *parsers*
 - Examples: arithmetic expressions, most programming languages
- Type 1 – Context-Sensitive
 - Rules may depend on context
 - Recognized by *linear bounded automata*
 - Examples: Some programming language constraints (e.g., type checking)
- Type 0 – Recursively Enumerable
 - Most general class of computable languages
 - Recognizable by Turing machines
 - Examples: Any program that can be executed



Example: A Language of Valid Identifiers

- $\Sigma = \{a, b, c, \dots, y, z, A, \dots, X, 0, 1, \dots, 9\}$
- $\Sigma^* = \{\varepsilon, a, aa, ab, aaa, \dots, a0, a00, a01, \dots, 0, 00, 01, \dots, 0a, 0aa, \dots\}$
- $L = \{a, aa, ab, aaa, \dots, a0, a00, a01, \dots, \}$
- The language L is *infinite*, but it does not allow all words from Σ^*

- We cannot enumerate all elements of the language, we need to define a recognizer that answers for a given word yes/no
- This is an example of a regular language, it only restricts how sequences of symbols can appear

Example: A Language of Paired Parenthesis

- $\Sigma = \{ (,) \}$
- $\Sigma^* = \{ \varepsilon, (, ((, (((, \dots, (, ()), (()), \dots, ()(), ((())(), \dots, (((())(), \dots \}$
- $L = \{ \varepsilon, (), (()), \dots, ()(), ((())(), \dots \}$
- This is an example of a context-free language, it is not regular because it must remember how many parentheses are currently open
- As languages become more expressive, recognizing them requires more resources

✓ Example: A Non-Context-Free Language

- $\Sigma = \{ a, b, c \}$
- $L = \{ \varepsilon, abc, aabbcc, aaabbbccc, \dots \}$
- This is an example of a language that is not context-free, recognition requires memory beyond a single stack
- We will not address the recognition of these languages in this course

-- Nuno Macedo